

BUSINESS REQUIREMENTS DOCUMENT

Online Claims Pre-Submission Validation Initiative

Claims Processing Optimization — Health Insurance Claims Denial Reduction

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Version	1.0
Status	Final — Approved for Development
Prepared By	Snega M, Business Analyst
Business Sponsor	VP, Claims Operations
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Revision History

Version	Date	Author	Description of Change
0.1	01-Jul-2026	Snega M	Initial draft based on RCA findings
0.2	05-Jul-2026	Snega M	Incorporated stakeholder feedback from review session
1.0	09-Jul-2026	Snega M	Finalized and approved for development handoff

1. Executive Summary

Analysis of 4,500 health insurance claims identified that General Practice claims submitted through the online channel deny at a rate of 39.2%, nearly six points above the 33.6% portfolio-wide average. Root cause analysis traced this to the absence of a pre-submission validation step confirming that a claim's procedure code is consistent with its diagnosis code.

This document defines the business requirements for introducing real-time, specialty-specific validation on the online claims portal. The initiative is expected to reduce the General Practice online denial rate to the portfolio average or below within two quarters, reducing rework for Claims Operations and improving the provider submission experience.

2. Business Objectives

Objective	Description
Primary	Reduce General Practice online-channel claim denial rate from 39.2% to $\leq 33.6\%$ within two quarters of release
Secondary	Establish a recurring denial-pattern reporting process to detect future risk combinations proactively
Secondary	Improve provider satisfaction by surfacing actionable errors at point of submission rather than after payer review

3. Project Scope

In Scope

- Real-time validation rule engine for diagnosis-to-procedure code pairing on the online claims portal
- Specialty-specific validation logic, prioritized for General Practice at launch (Phase 1)
- Provider-facing inline error messaging with correction guidance
- Validation failure logging for downstream analysis
- Monthly automated denial-pattern report by specialty and submission channel

Out of Scope

- Changes to the Paper or Phone submission channels
- Retroactive re-review of previously denied or closed claims
- Changes to payer-side adjudication rules outside the intake portal
- Validation logic for specialties other than General Practice (deferred to Phase 2)

4. Stakeholders & RACI

Stakeholder	Role	R	A	C	I
VP, Claims Operations	Business Sponsor		X		
Business Analyst (Snega M)	Requirements Owner	X			
Product Manager	Solution Owner	X			
Engineering Lead	Technical Delivery	X			
Compliance	Coding Standard Approval			X	
Provider Relations	Provider Communication			X	
QA / UAT Team	Test & Validation	X			

Stakeholder	Role	R	A	C	I
Claims Processing Team	End User				X

R = Responsible, A = Accountable, C = Consulted, I = Informed

5. Current State (AS-IS)

Claims are submitted through three channels — Online, Paper, and Phone. Paper and Phone submissions pass through a staff intermediary who performs an informal completeness check; the Online channel allows direct, unassisted provider self-submission with no equivalent check. There is no system-enforced validation of diagnosis-to-procedure code consistency on any channel. Denial reasons are reviewed by Claims Operations on an ad hoc basis, with no recurring cadence feeding findings back into portal requirements.

6. Future State (TO-BE)

Dimension	Current State	Future State
Validation	No pre-submission check; errors caught only after payer review	Real-time diagnosis-to-procedure code validation before submission
Denial Visibility	Ad hoc review, no recurring cadence	Monthly automated denial-pattern report by specialty and channel
Provider Experience	Denial learned days after submission	Actionable error shown at point of entry, correctable in-session
GP Online Denial Rate	39.2%	Target: ≤33.6%

7. Functional Business Requirements

ID	Requirement	Priority
BR-01	The system shall validate that the submitted procedure code is clinically consistent with the submitted diagnosis code before allowing claim submission on the online channel.	Must Have
BR-02	The system shall apply specialty-specific validation rules, beginning with General Practice, without altering the intake flow for other specialties.	Must Have
BR-03	The system shall display an inline, plain-language error message identifying the specific field causing a validation failure.	Must Have
BR-04	The system shall allow a provider to correct and resubmit a flagged claim without restarting the entire submission.	Should Have
BR-05	The system shall log all validation failures with specialty, channel, and reason code for downstream reporting.	Must Have
BR-06	The system shall generate a monthly denial-pattern report, segmented by specialty and submission channel, for the Claims Operations and Product teams.	Should Have
BR-07	The system shall not increase average online claim submission time by more than 15 seconds versus current baseline.	Must Have

8. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	Validation response time shall not exceed 2 seconds per submission under normal load.
NFR-02	Availability	The validation service shall maintain 99.5% uptime during business hours.
NFR-03	Resilience	If the validation service is unavailable, the system shall fail gracefully and route the claim to manual review rather than blocking submission.
NFR-04	Auditability	All validation decisions shall be logged and retained for a minimum of 12 months for audit purposes.
NFR-05	Usability	Error messages shall be understandable to a non-technical provider without support-desk assistance.

9. Business Rules

- BRUL-01: A claim shall not be submitted on the online channel if its procedure code is not present in the approved pairing list for its diagnosis code.
- BRUL-02: Validation rules apply only to the specialty explicitly enabled for the current rollout phase (General Practice in Phase 1).
- BRUL-03: A claim that fails validation is not counted as "submitted" until it passes validation and is accepted by the system.
- BRUL-04: Denial-pattern reports must be generated on the first business day of each month for the preceding calendar month.

10. Assumptions

- Diagnosis and procedure code reference tables (ICD/CPT-equivalent) are available and maintained by Compliance.

- Engineering capacity allows delivery within a single quarter; General Practice rollout is Phase 1, with remaining specialties in Phase 2.
- Existing Power BI / SQL reporting infrastructure can ingest the new validation failure log format without material rework.

11. Constraints

- Validation logic will not be applied retroactively to claims already in Pending or Denied status.
- The Paper and Phone submission channels are out of scope and must remain functionally unchanged.
- Solution must operate within the existing online portal's current technology stack; no new platform migration is in scope.

12. Risks & Mitigations

Risk	Impact	Mitigation
Validation rules incorrectly reject legitimate claims	High	Phase 1 limited to General Practice; monitor false-positive rate closely during first 30 days post-release
Reference code tables become outdated	Medium	Compliance to own a quarterly review cadence for code-pairing tables
Increased submission time discourages online use	Medium	NFR-01/BR-07 performance guardrails; monitor submission time weekly post-release
Provider confusion over new error messages	Low	Provider Relations to issue rollout communication ahead of release; UAT includes provider-readability review

13. Dependencies

- Availability of the diagnosis-to-procedure code reference dataset from Compliance prior to development start
- Engineering capacity allocation confirmed for the target quarter
- Existing denial-reason logging pipeline extended to support new validation failure fields

14. Success Metrics / KPIs

Metric	Baseline	Target
GP Online Denial Rate	39.2%	≤ 33.6%
Portfolio-wide Denial Rate	33.6%	≤ 30%
Avg. Submission Time (Online)	Current baseline	+15 sec or less
Denial Report Cadence	Ad hoc	Monthly, automated
False-Positive Validation Rate	N/A (new metric)	< 2% of GP online submissions

15. Glossary

Term	Definition
Denial Rate	Percentage of submitted claims resulting in a Denied status

Term	Definition
Pre-submission Validation	A rule-based check performed before a claim is accepted into the processing queue
Diagnosis-to-Procedure Pairing	Business rule confirming a procedure code is clinically valid for a given diagnosis code
UAT	User Acceptance Testing — validation performed by business/QA stakeholders before release
RACI	Responsible, Accountable, Consulted, Informed — a stakeholder responsibility matrix

16. Approval & Sign-off

By signing below, stakeholders confirm they have reviewed this Business Requirements Document and agree it accurately reflects the business need and requirements for this initiative.

VP, Claims Operations (Sponsor)	_____ Date: _____
Product Lead	_____ Date: _____
Compliance Lead	_____ Date: _____
Engineering Lead	_____ Date: _____